BioInf – sample test 1 questions

This document shows the type of questions that might appear in the first test. It does not imply that only this type of questions will appear in the test or that the test will follow this structure. The actual test may have a different number of questions and cover other topics that were discussed during the classes.

GROUP I (Lectures 1-4)

1. Show the open reading frames (ORFs) of this DNA sequence, without translating to aminoacids: `CTAATCGG`
2. Given a DNA sequence, a codon translation table and an aminoacid, explain how we can determine the codon bias for that codon
3. Draw the #G-#C skew diagram of the following DNA sequence; where would you assume replication starts? `ACGCCACATGGGTG`

GROUP II (Lectures 5-8)

1. In the following questions consider the alignment parameters: match: +1, mismatch: −1, gap penalty = −1.

   Consider the application of the Needleman-Wunsch algorithm to the DNA sequences for global pairwise sequence alignment:

   ```
   GATTG
   TGTA
   ```

   a. Compute the S and T matrices, the optimal alignment and its score. Show all alternative optimal alignments.
   b. Would the best alignment score be higher, lower, or equal if we considered a semi-global alignment instead? Justify.
2. Considering the BLAST algorithm for sequence database search, as implemented by NCBI:
   a. What is the difference between BLASTN and BLASTP, and in what scenarios would each be used?
3. Explain the word neighborhood generation of the BLAST algorithm and how it fits with the full algorithm.
4. What is the time and memory complexity of the Needleman-Wunsch algorithm for pairwise sequence alignment and for multiple sequence alignment, in both cases considering all sequences have the same length?
5. Calculate the sum-of-pairs score and entropy score of the following alignment. Use the following parameters: gap penalty = -1, match = 1, mismatch = -1. For the entropy score, you do not have to calculate the final value.

   ```
   AATG
   -CTG
   A-TG
   ```

GROUP III (Lecture 9-12)

1. Consider the multiple sequence alignment of DNA sequences 1, 2, 3, 4, 5:

```
1: TAACCAATGG
2: AAAGCCCTGT
3: AACTCCGTCG
4: AACTGAGTCT
5: ACCTGAGTCT
```

a. Compute the pairwise number of mismatches between the taxa and write down the corresponding distance matrix.

b. Use the UPGMA algorithm to compute the phylogenetic tree from the distance matrix. Show all the distance matrices and trees computed during the algorithm.

c. Calculate the distance in the tree between every pair of sequences.

d. Which pairs of sequences show the greatest difference between their distances represented in the obtained phylogenetic tree and their distances represented in the matrix?

2. When implementing a genetic algorithm for multiple sequence alignment, how can we calculate the fitness of an individual?

3. Suppose that we use branch-and-bound to discover a motif of **length 3** in the set of sequences of **GROUP III**. Consider the internal node represented by [3,0,2] where only the starting positions of sequences 1, 2, 3 are defined, respectively 3, 0 and 2 (i.e. for sequences 4 and 5 any starting position is still possible). Compute the upper bound of the score function for this node, **assuming that the score is the sum of count of most frequent symbol of each column**. If you already have found an initial candidate solution [0,0,0,0,0] would you discard this internal node? Justify.

4. Suppose that we use the heuristic consensus algorithm to discover a motif of **length 4** in the same set of sequences of **GROUP III**. It starts by considering only the first two sequences and finds their initial positions with the best score: both sequences starting at 1 ([1,1]). How would the algorithm proceed to compute the initial positions of the next sequences? What would be the starting position of sequence 3? Justify.

5. Suppose that we use the Gibbs sampling algorithm to discover a motif of **length 5** in the same set of sequences of **GROUP III**. It starts by randomly selecting the initial positions along the input sequences.
    a. If the obtained initial positions are [4,5,2,4,0], compute the initial score given by sum of the number of mismatches between each motif and the consensus.
    b. Then, in the next step of Gibbs algorithm, a new profile matrix is computed after excluding a random sequence. Assume that sequence 2 is selected and compute the new profile matrix.

c. For each position p in sequence 2, calculate the probability of the sub-sequence started in p with length 5 being generated by the new profile matrix